

Factor Models, PCA and the Fama-French Three-Factor Model

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Abstract

Using daily returns on 22 large US stocks from January 2020 to December 2024, we compare a statistical factor model built from principal components with the Fama-French three-factor model (FF3). We choose the number of principal components from the scree plot, compare both models on fit and residual correlation for a big-tech reference group (AAPL, MSFT, AMZN, GOOGL, META, NVDA), measure how well each model reproduces the sample covariance matrix, and relate the first three principal components to the Fama-French factors. Everything is implemented in R.

1 Data

The dataset contains 1258 daily returns (in %) from 2 January 2020 to 31 December 2024 on 22 stocks, of which the first six form the big-tech reference group. It also contains the excess market return R_M^e , the risk-free rate, and the SMB and HML factors. Stock returns come from Yahoo Finance; the factors and risk-free rate come from Kenneth French's data library.

```
Dataset      <- read.csv("data/DataAssignment2.csv", check.names = FALSE)
ExcessMarket <- Dataset[, 24]
RiskFree     <- Dataset[, 25]
SMB          <- Dataset[, 26]
HML          <- Dataset[, 27]
Returns      <- as.matrix(Dataset[, 2:23])
t <- nrow>Returns); N <- ncol>Returns)      # 1258 and 22
```

2 How many principal components?

We run PCA on the sample covariance matrix of the 22 return series. All series are in the same units, so we do not standardise them. The share of total variance explained by component k is $\ell_k / \sum_j \ell_j$, where ℓ_k is the k -th eigenvalue. We choose the number of factors K whose cumulative share is closest to 90%.

```
PCAResult      <- prcomp>Returns, center = TRUE, scale. = FALSE)
VarExplained   <- PCAResult$sdev^2 / sum(PCAResult$sdev^2)
CumVarExplained <- cumsum(VarExplained)
K <- which.min(abs(CumVarExplained - 0.90)) # K = 11
```

The first component explains 46.4% of the total variance and the second a further 14.4%, after which the individual contributions flatten out quickly. Ten components explain 88.85% and eleven explain 90.56%. Eleven is closest to 90%, so we use $K = 11$.

Needing half as many factors as stocks to reach 90% shows that a large part of daily return variation is stock-specific. The first component is a strong common factor, but the remaining variance is spread thinly over many directions.

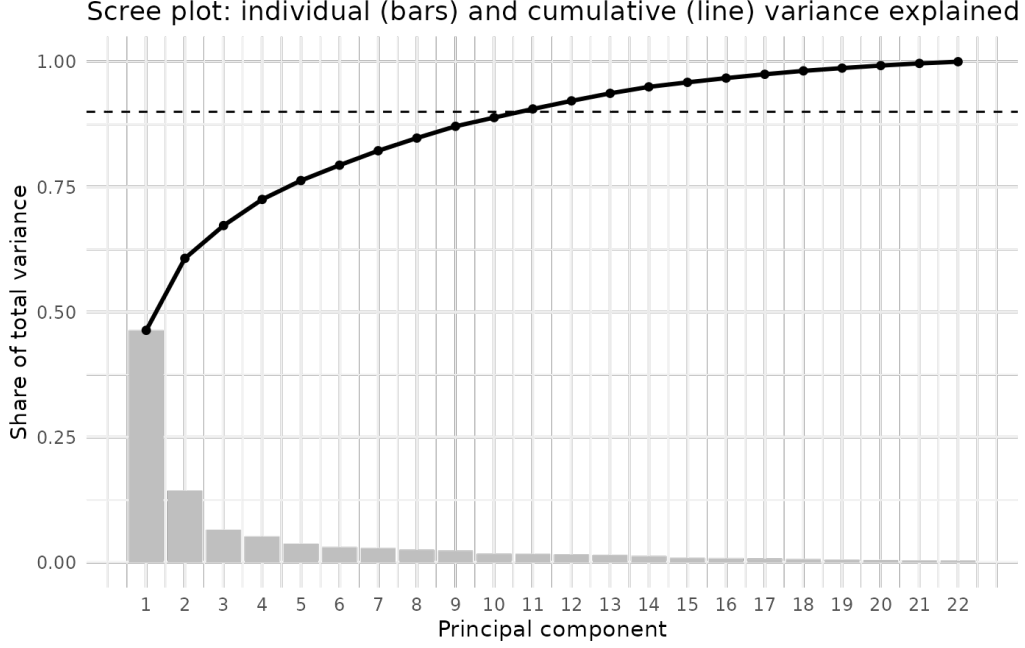


Figure 1: Scree plot. The dashed line marks 90%.

3 PCR(11) versus FF3

For each stock i we estimate

$$\text{PCR(11): } R_{i,t} = \alpha_i + \sum_{k=1}^{11} \beta_{ik} PC_{k,t} + \varepsilon_{i,t}, \quad (1)$$

$$\text{FF3: } R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,M} R_{M,t}^e + \beta_{i,S} SMB_t + \beta_{i,H} HML_t + \varepsilon_{i,t}. \quad (2)$$

For FF3 we use excess stock returns, as the model is specified in excess returns. Because the daily risk-free rate is close to constant, this has almost no effect on slopes, residuals or covariances.

```
Factors1 <- PCAResult$x[, 1:K]
ExcessReturns <- Returns - RiskFree
PCROut1 <- lm>Returns ~ Factors1)
FF3Out1 <- lm(ExcessReturns ~ ExcessMarket + SMB + HML)

AdjR2_PCR <- sapply(summary(PCROut1), function(x) x$adj.r.squared)
AdjR2_FF3 <- sapply(summary(FF3Out1), function(x) x$adj.r.squared)
AdjR2_PCR_BT <- mean(AdjR2_PCR[1:6]); AdjR2_FF3_BT <- mean(AdjR2_FF3[1:6])

ResidCorr_PCR_BT <- cor(PCROut1$residuals[, 1:6])
ResidCorr_FF3_BT <- cor(FF3Out1$residuals[, 1:6])
AvgAbsResidCorr_PCR_BT <- mean(abs(ResidCorr_PCR_BT[upper.tri(ResidCorr_PCR_BT)]))
AvgAbsResidCorr_FF3_BT <- mean(abs(ResidCorr_FF3_BT[upper.tri(ResidCorr_FF3_BT)]))
```

Big-tech group	PCR(11)	FF3
Average adjusted R^2	0.914	0.611
Average absolute off-diagonal residual correlation	0.321	0.096

Table 1: Fit of the two models for the six big-tech stocks.

The two measures point in opposite directions.

	AAPL	MSFT	AMZN	GOOGL	META	NVDA
Adj. R^2 PCR(11)	0.808	0.833	0.971	0.874	0.999	0.999
Adj. R^2 FF3	0.677	0.765	0.558	0.597	0.458	0.610

Table 2: Adjusted R^2 per big-tech stock.

Fit. PCR(11) explains far more of each big-tech stock’s variation (0.914 against 0.611). Part of this is mechanical. The principal components are built from these same returns, in sample, and PCA on the covariance matrix gives most weight to the most volatile stocks. META and NVDA have the largest variances in the sample, so they almost span their own components, which is why their adjusted R^2 is 0.999.

Residual correlation. A factor model assumes the residuals are uncorrelated across stocks. FF3 comes much closer to this (0.096 against 0.321). Its remaining residual correlations are mostly small and positive, the largest being between GOOGL, META, AMZN and MSFT (around 0.20 to 0.26). This is common tech exposure that the market, size and value factors do not capture. The PCR residuals are strongly correlated, often negatively (for example -0.68 between AAPL and GOOGL). With 11 of 22 components removed, each residual is a combination of the 11 remaining components, so the residuals share a low-dimensional structure by construction.

In short, PCR(11) fits each stock better in sample, but FF3 better satisfies the structural assumption of a factor model for this group.

4 Implied covariance matrices

Under a factor model with loadings B , factor covariance Σ_F and diagonal idiosyncratic covariance D , the implied covariance matrix of returns is

$$\Sigma_R = B\Sigma_F B^\top + D. \quad (3)$$

We estimate D from the residual variances with a degrees-of-freedom correction, $\frac{t-1}{t-K-1}$ for PCR and $\frac{t-1}{t-4}$ for FF3. We then compare the 6×6 big-tech block of each implied matrix with the sample covariance matrix using the Frobenius distance $\|A - S\|_F = (\sum_{i,j} (a_{ij} - s_{ij})^2)^{1/2}$.

```
ResCovPCR <- (t - 1) / (t - K - 1) * diag(diag(cov(PCROut1$residuals)))
ResCovFF3 <- (t - 1) / (t - 4) * diag(diag(cov(FF3Out1$residuals)))
BetaPCR <- t(coef(PCROut1)[-1, ])
BetaFF3 <- t(coef(FF3Out1)[-1, ])
FF3Fact1 <- cbind(ExcessMarket, SMB, HML)

CovRetPCR <- BetaPCR %*% cov(Factors1) %*% t(BetaPCR) + ResCovPCR
CovRetFF3 <- BetaFF3 %*% cov(FF3Fact1) %*% t(BetaFF3) + ResCovFF3
SampleCov <- cov>Returns)

FrobPCR <- norm(CovRetPCR[1:6, 1:6] - SampleCov[1:6, 1:6], type = "F")
FrobFF3 <- norm(CovRetFF3[1:6, 1:6] - SampleCov[1:6, 1:6], type = "F")
```

PCR(11) gives the better covariance fit, with a Frobenius distance less than half that of FF3. In both models the diagonal matches the sample variances almost exactly, because the idiosyncratic variances absorb whatever the factors miss. The difference therefore comes from the covariances.

FF3 systematically underestimates the covariances among the tech stocks, for example MSFT and GOOGL (2.648 against 2.935) and AMZN and META (3.225 against 3.904). This is the positive residual correlation from the previous section: FF3 assumes D is diagonal and so

PCR(11) implied						
	AAPL	MSFT	AMZN	GOOGL	META	NVDA
AAPL	3.990	3.051	2.678	3.084	3.124	4.112
MSFT	3.051	3.696	3.018	3.146	3.362	4.502
AMZN	2.678	3.018	5.135	3.086	3.903	4.427
GOOGL	3.084	3.146	3.086	4.196	3.759	4.115
META	3.124	3.362	3.903	3.759	7.997	5.136
NVDA	4.112	4.502	4.427	4.115	5.136	11.513

FF3 implied						
	AAPL	MSFT	AMZN	GOOGL	META	NVDA
AAPL	3.987	2.756	2.700	2.600	3.118	4.271
MSFT	2.756	3.693	2.780	2.648	3.191	4.377
AMZN	2.700	2.780	5.139	2.608	3.225	4.483
GOOGL	2.600	2.648	2.608	4.195	3.011	4.129
META	3.118	3.191	3.225	3.011	8.005	5.072
NVDA	4.271	4.377	4.483	4.129	5.072	11.521

Sample						
	AAPL	MSFT	AMZN	GOOGL	META	NVDA
AAPL	3.983	2.870	2.680	2.653	3.179	4.107
MSFT	2.870	3.690	2.954	2.935	3.368	4.447
AMZN	2.680	2.954	5.134	3.004	3.904	4.458
GOOGL	2.653	2.935	3.004	4.191	3.712	4.128
META	3.179	3.368	3.904	3.712	7.997	5.138
NVDA	4.107	4.447	4.458	4.128	5.138	11.513

Table 3: Big-tech covariance submatrices (daily returns in %, so units are %²).

	PCR(11)	FF3
Frobenius distance to sample covariance	0.753	1.619

Table 4: Covariance fit for the big-tech group.

drops the shared tech exposure. PCR(11) has larger residual correlations, but its residual variances are much smaller, so ignoring them costs little in covariance terms. Its largest errors are overestimates, such as AAPL and GOOGL (3.084 against 2.653).

One caveat applies. PCR(11) uses 11 factors against 3 and is estimated on the same sample it is evaluated on, so it is favoured in sample. An out-of-sample comparison would give a fairer test.

5 Principal components and the Fama-French factors

```
PCFactors3 <- PCAResult$x[, 1:3]
CorrPCFF <- cor(PCFactors3, cbind(MktRF = ExcessMarket, SMB = SMB, HML = HML))
R2_FF_on_PC <- sapply(list(ExcessMarket, SMB, HML),
  function(f) summary(lm(f ~ PCFactors3))$r.squared)
```

The sign of a principal component is arbitrary, so only the absolute size of each correlation matters.

PC1 is the market factor. Its correlation with R_M^e is -0.972 . All 22 stocks load on it

	R_M^e	SMB	HML
PC1	−0.972	−0.130	0.067
PC2	−0.033	−0.037	−0.801
PC3	−0.021	0.269	0.199
R^2 on PC1 to PC3	0.945	0.090	0.685

Table 5: Correlations between the first three principal components and the Fama-French factors. The last row is the R^2 from regressing each factor on the three components jointly.

with the same sign, so it is a broad market portfolio.

PC2 is a value factor. Its correlation with HML is -0.801 . The loadings show why: PC2 loads positively on the big-tech growth stocks (NVDA, META, AMZN) and negatively on value stocks such as XOM, CVX, BAC and JPM. It is effectively a growth-minus-value portfolio, the mirror image of HML.

PC3 is not clearly any FF factor. Its correlations are weak (0.269 with SMB and 0.199 with HML). Its loadings contrast NVDA, energy and banks with defensive stocks such as PG, PFE, PEP, WMT and JNJ, so it looks more like a cyclical versus defensive factor.

SMB is not captured. The three components jointly explain 94.5% of the variance of the market factor and 68.5% of HML, but only 9.0% of SMB. This is expected: all 22 stocks are mega-caps, so the sample contains no small-cap variation from which a size factor could emerge. HML is captured reasonably well, because the sample contains both growth (tech) and value (energy, banks) stocks.

6 Conclusions

- Reaching 90% of the variance of 22 large-cap stocks takes 11 principal components, although the first one alone explains 46%.
- PCR(11) fits the big-tech stocks better and reproduces their covariance matrix more closely, but its residuals are strongly correlated. FF3 fits worse but leaves only weak residual correlation, mainly a shared tech component.
- The first two components correspond to the market and value factors. A size factor does not appear, because the sample contains no small firms.